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Assignment 3

QUESTION 1

Image to generate (4×4 grid):

p_{11}	p_{12}	p_{13}	p_{14}
p_{21}	p_{22}	p_{23}	p_{24}
p_{31}	p_{32}	p_{33}	p_{34}
p_{41}	p_{42}	p_{43}	p_{44}

Masked Conv of Type A → Kernel:

1	-3	1
2	-1	2
-1	-2	1

Activation: ReLU

Logits: $w = [0.2, 0.3, 0.1, 0.4]$, $b = [0.1, -0.2, 0.3, -0.1]$

Softmax with temperature T: $\hat{p}_i = \exp(L_i/T) / \Sigma \exp(L_i/T)$

Pixel [1,1]

Image Patch: Zero-padded — center at p_{11}, all visible pixels are 0

Convolution result: $0 \times \text{kernel} = 0$

h = ReLU = 0

Logits (z = w·h + b):

$z_0 = 0.2 \times 0 + 0.1 = 0.1$

$z_1 = 0.3 \times 0 - 0.2 = -0.2$

$z_2 = 0.1 \times 0 + 0.3 = 0.3$

$z_3 = 0.4 \times 0 - 0.1 = -0.1$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
0.1	1.1052	0.2645	1.0513	0.2516	1.2214	0.2923
-0.2	0.8187	0.1959	0.9048	0.2165	0.6703	0.1604
0.3	1.3499	0.3230	1.1618	0.2780	1.8221	0.4361
-0.1	0.9048	0.2165	0.9512	0.2276	0.8187	0.1959
Sum	4.1786	1.0	4.0692	1.0	4.5326	1.0

Logit Selected = 2

Generated Image:

2	$?_{12}$	$?_{13}$	$?_{14}$
$?_{21}$	$?_{22}$	$?_{23}$	$?_{24}$
$?_{31}$	$?_{32}$	$?_{33}$	$?_{34}$
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [1,2]

Image Patch: Left neighbor = $p_{11} = 2$; patch dot kernel = 4

Convolution result: $(2\times2) + (0\times\text{rest}) = 4$

h = ReLU = 4

Logits (z = w·h + b):

$z_0 = 0.2 \times 4 + 0.1 = 0.9$

$z_1 = 0.3 \times 4 - 0.2 = 1.0$

$z_2 = 0.1 \times 4 + 0.3 = 0.7$

$z_3 = 0.4 \times 4 - 0.1 = 1.5$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
0.9	2.4596	0.2107	1.5683	0.2322	6.0496	0.1610
1.0	2.7183	0.2329	1.6487	0.2441	7.3891	0.1966
0.7	2.0138	0.1725	1.4191	0.2101	4.0552	0.1079
1.5	4.4817	0.3839	2.1170	0.3135	20.0855	0.5345
Sum	11.6733	1.0	6.7531	1.0	37.5794	1.0

Logit Selected = 3

Generated Image:

2	3	$?_{13}$	$?_{14}$
$?_{21}$	$?_{22}$	$?_{23}$	$?_{24}$
$?_{31}$	$?_{32}$	$?_{33}$	$?_{34}$
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [1,3]

Image Patch: Left two neighbors = 2,3; patch dot kernel = 6

Convolution result: $(3\times2) + (0\times\text{rest}) = 6$

h = ReLU = 6

Logits (z = w·h + b):

$z_0 = 0.2 \times 6 + 0.1 = 1.3$

$z_1 = 0.3 \times 6 - 0.2 = 1.6$

$z_2 = 0.1 \times 6 + 0.3 = 0.9$

$z_3 = 0.4 \times 6 - 0.1 = 2.3$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
1.3	3.6693	0.1743	1.9155	0.2160	13.4637	0.0938
1.6	4.9530	0.2352	2.2255	0.2510	24.5325	0.1709
0.9	2.4596	0.1168	1.5683	0.1769	6.0496	0.0421
2.3	9.9742	0.4737	3.1582	0.3562	99.4843	0.6931
Sum	21.0561	1.0	8.8676	1.0	143.530	1.0

Logit Selected = 3

Generated Image:

2	3	3	$?_{14}$
$?_{21}$	$?_{22}$	$?_{23}$	$?_{24}$
$?_{31}$	$?_{32}$	$?_{33}$	$?_{34}$
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [1,4]

Image Patch: Same patch as [1,3] due to boundary padding

Convolution result: $(3\times2) + (0\times\text{rest}) = 6$

h = ReLU = 6

Logits (z = w·h + b):

$z_0 = 0.2 \times 6 + 0.1 = 1.3$

$z_1 = 0.3 \times 6 - 0.2 = 1.6$

$z_2 = 0.1 \times 6 + 0.3 = 0.9$

$z_3 = 0.4 \times 6 - 0.1 = 2.3$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
1.3	3.6693	0.1743	1.9155	0.2160	13.4637	0.0938
1.6	4.9530	0.2352	2.2255	0.2510	24.5325	0.1709

0.9	2.4596	0.1168	1.5683	0.1769	6.0496	0.0421
2.3	9.9742	0.4737	3.1582	0.3562	99.4843	0.6931
Sum	21.0561	1.0	8.8676	1.0	143.530	1.0

Logit Selected = 3

Generated Image:

2	**3**	**3**	**3**
$?_{21}$ $?_{31}$	$?_{22}$ $?_{32}$	$?_{23}$ $?_{33}$	$?_{24}$ $?_{34}$
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [2,1]

Image Patch: Row above = [0,2,3,3]; ReLU clips negative to 3

Convolution result: $(0\times1)+(2\times-3)+(3\times1)+(0\times2)+... = -3 \rightarrow \text{ReLU}(-3) = 3$

h = ReLU = 3

Logits (z = w·h + b):

$z_0 = 0.2 \times 3 + 0.1 = 0.7$

$z_1 = 0.3 \times 3 - 0.2 = 0.7$

$z_2 = 0.1 \times 3 + 0.3 = 0.6$

$z_3 = 0.4 \times 3 - 0.1 = 1.1$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.6	1.8221	0.2058	1.3499	0.2280	3.3201	0.1623
1.1	3.0042	0.3393	1.7333	0.2927	9.0250	0.4412
Sum	8.8538	1.0	5.9212	1.0	20.4555	1.0

Logit Selected = 3

Generated Image:

2	**3**	**3**	**3**
3	$?_{22}$	$?_{23}$	$?_{24}$
$?_{31}$	$?_{32}$	$?_{33}$	$?_{34}$
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [2,2]

Image Patch: Top-row context [2,3,3] + left=3

Convolution result: $(2\times1)+(3\times-3)+(3\times1)+(3\times2) = 2$

h = ReLU = 2

Logits (z = w·h + b):

$z_0 = 0.2 \times 2 + 0.1 = 0.5$

$z_1 = 0.3 \times 2 - 0.2 = 0.4$

$z_2 = 0.1 \times 2 + 0.3 = 0.5$

$z_3 = 0.4 \times 2 - 0.1 = 0.7$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
0.5	1.6487	0.2424	1.2840	0.2465	2.7183	0.2320
0.4	1.4918	0.2193	1.2214	0.2345	2.2255	0.1899
0.5	1.6487	0.2424	1.2840	0.2465	2.7183	0.2320
0.7	2.0138	0.2960	1.4191	0.2725	4.0552	0.3461
Sum	6.8030	1.0	5.2085	1.0	11.7173	1.0

Logit Selected = 3

Generated Image:

2	3	3	3
3	3	$?_{23}$	$?_{24}$
$?_{31}$	$?_{32}$	$?_{33}$	$?_{34}$
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [2,3]

Image Patch: Top-row [3,3,3] + left=3

Convolution result: $(3\times1)+(3\times-3)+(3\times1)+(3\times2) = 3$

h = ReLU = 3

Logits (z = w·h + b):

$z_0 = 0.2 \times 3 + 0.1 = 0.7$

$z_1 = 0.3 \times 3 - 0.2 = 0.7$

$z_2 = 0.1 \times 3 + 0.3 = 0.6$

$z_3 = 0.4 \times 3 - 0.1 = 1.1$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.6	1.8221	0.2058	1.3499	0.2280	3.3201	0.1623
1.1	3.0042	0.3393	1.7333	0.2927	9.0250	0.4412
Sum	8.8538	1.0	5.9212	1.0	20.4555	1.0

Logit Selected = 3

Generated Image:

2 3 $?_{31}$	3 3 $?_{32}$	3 3 $?_{33}$	3 $?_{24}$ $?_{34}$
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [2,4]

Image Patch: Right boundary padding causes sum = 0

Convolution result: $(3\times1)+(3\times-3)+(0)+(3\times2) = 0$

h = ReLU = 0

Logits (z = w·h + b):

$z_0 = 0.2 \times 0 + 0.1 = 0.1$

$z_1 = 0.3 \times 0 - 0.2 = -0.2$

$z_2 = 0.1 \times 0 + 0.3 = 0.3$

$z_3 = 0.4 \times 0 - 0.1 = -0.1$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
0.1	1.1052	0.2645	1.0513	0.2516	1.2214	0.2923
-0.2	0.8187	0.1959	0.9048	0.2165	0.6703	0.1604
0.3	1.3499	0.3230	1.1618	0.2780	1.8221	0.4361
-0.1	0.9048	0.2165	0.9512	0.2276	0.8187	0.1959
Sum	4.1786	1.0	4.0692	1.0	4.5326	1.0

Logit Selected = 2

Generated Image:

2 3 $?_{31}$	3 3 $?_{32}$	3 3 $?_{33}$	3 2 $?_{34}$
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [3,1]

Image Patch: Row2 = [0,3,3,3]; ReLU converts −6 to 6

Convolution result: $(0\times1)+(3\times-3)+(3\times1) = -6 \rightarrow \text{ReLU}(-6) = 6$

h = ReLU = 6

Logits (z = w·h + b):

$z_0 = 0.2 \times 6 + 0.1 = 1.3$

$z_1 = 0.3 \times 6 - 0.2 = 1.6$

$z_2 = 0.1 \times 6 + 0.3 = 0.9$

$z_3 = 0.4 \times 6 - 0.1 = 2.3$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
1.3	3.6693	0.1743	1.9155	0.2160	13.4637	0.0938
1.6	4.9530	0.2352	2.2255	0.2510	24.5325	0.1709
0.9	2.4596	0.1168	1.5683	0.1769	6.0496	0.0421
2.3	9.9742	0.4737	3.1582	0.3562	99.4843	0.6931
Sum	21.0561	1.0	8.8676	1.0	143.530	1.0

Logit Selected = 3

Generated Image:

2 3 3	3 3 $?_{32}$	3 3 $?_{33}$	3 2 $?_{34}$
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [3,2]

Image Patch: Patch top-row [3,3,3] + left=3

Convolution result: $(3\times1)+(3\times-3)+(3\times1)+(3\times2) = 3$

h = ReLU = 3

Logits ($z = w \cdot h + b$):

$z_0 = 0.2 \times 3 + 0.1 = 0.7$

$z_1 = 0.3 \times 3 - 0.2 = 0.7$

$z_2 = 0.1 \times 3 + 0.3 = 0.6$

$z_3 = 0.4 \times 3 - 0.1 = 1.1$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.6	1.8221	0.2058	1.3499	0.2280	3.3201	0.1623
1.1	3.0042	0.3393	1.7333	0.2927	9.0250	0.4412
Sum	8.8538	1.0	5.9212	1.0	20.4555	1.0

Logit Selected = 3

Generated Image:

2 3	3 3	3 3	3 2
3	3	$?_{33}$	$?_{34}$
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [3,3]

Image Patch: Patch top-row [3,3,2] + left=3

Convolution result: $(3\times1)+(3\times-3)+(2\times1)+(3\times2) = 2$

h = ReLU = 2

Logits ($z = w \cdot h + b$):

$z_0 = 0.2 \times 2 + 0.1 = 0.5$

$z_1 = 0.3 \times 2 - 0.2 = 0.4$

$z_2 = 0.1 \times 2 + 0.3 = 0.5$

$z_3 = 0.4 \times 2 - 0.1 = 0.7$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
0.5	1.6487	0.2424	1.2840	0.2465	2.7183	0.2320
0.4	1.4918	0.2193	1.2214	0.2345	2.2255	0.1899
0.5	1.6487	0.2424	1.2840	0.2465	2.7183	0.2320
0.7	2.0138	0.2960	1.4191	0.2725	4.0552	0.3461

Sum	6.8030	1.0	5.2085	1.0	11.7173	1.0

Logit Selected = 3

Generated Image:

2	3	3	3
3	3	3	2
3	3	3	$?_{34}$
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [3,4]

Image Patch: Right edge + [3,2] patch top

Convolution result: $(3\times1)+(2\times-3)+(0)+(3\times2) = 3$

h = ReLU = 3

Logits (z = w·h + b):

$z_0 = 0.2 \times 3 + 0.1 = 0.7$

$z_1 = 0.3 \times 3 - 0.2 = 0.7$

$z_2 = 0.1 \times 3 + 0.3 = 0.6$

$z_3 = 0.4 \times 3 - 0.1 = 1.1$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.6	1.8221	0.2058	1.3499	0.2280	3.3201	0.1623
1.1	3.0042	0.3393	1.7333	0.2927	9.0250	0.4412
Sum	8.8538	1.0	5.9212	1.0	20.4555	1.0

Logit Selected = 3

Generated Image:

2	3	3	3
3	3	3	2
3	3	3	3
$?_{41}$	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [4,1]

Image Patch: Same boundary condition as [3,1]

Convolution result: $(0\times1)+(3\times-3)+(3\times1) = -6 \rightarrow \text{ReLU}(-6) = 6$

h = ReLU = 6

Logits (z = w·h + b):

$z_0 = 0.2 \times 6 + 0.1 = 1.3$

$z_1 = 0.3 \times 6 - 0.2 = 1.6$

$z_2 = 0.1 \times 6 + 0.3 = 0.9$

$z_3 = 0.4 \times 6 - 0.1 = 2.3$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
1.3	3.6693	0.1743	1.9155	0.2160	13.4637	0.0938
1.6	4.9530	0.2352	2.2255	0.2510	24.5325	0.1709
0.9	2.4596	0.1168	1.5683	0.1769	6.0496	0.0421
2.3	9.9742	0.4737	3.1582	0.3562	99.4843	0.6931
Sum	21.0561	1.0	8.8676	1.0	143.530	1.0

Logit Selected = 3

Generated Image:

2	3	3	3
3	3	3	2
3	3	3	3
3	$?_{42}$	$?_{43}$	$?_{44}$

Pixel [4,2]

Image Patch: Row3 = [3,3,3], left = 3

Convolution result: $(3\times1)+(3\times-3)+(3\times1)+(3\times2) = 3$

h = ReLU = 3

Logits (z = w·h + b):

$z_0 = 0.2 \times 3 + 0.1 = 0.7$

$z_1 = 0.3 \times 3 - 0.2 = 0.7$

$z_2 = 0.1 \times 3 + 0.3 = 0.6$

$z_3 = 0.4 \times 3 - 0.1 = 1.1$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5

0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.6	1.8221	0.2058	1.3499	0.2280	3.3201	0.1623
1.1	3.0042	0.3393	1.7333	0.2927	9.0250	0.4412
Sum	8.8538	1.0	5.9212	1.0	20.4555	1.0

Logit Selected = 3

Generated Image:

2	3	3	3
3	3	3	2
3	3	3	3
3	3	$?_{43}$	$?_{44}$

Pixel [4,3]

Image Patch: Same as [4,2]

Convolution result: $(3\times1)+(3\times-3)+(3\times1)+(3\times2) = 3$

h = ReLU = 3

Logits (z = w·h + b):

$z_0 = 0.2 \times 3 + 0.1 = 0.7$

$z_1 = 0.3 \times 3 - 0.2 = 0.7$

$z_2 = 0.1 \times 3 + 0.3 = 0.6$

$z_3 = 0.4 \times 3 - 0.1 = 1.1$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.7	2.0138	0.2274	1.4191	0.2397	4.0552	0.1982
0.6	1.8221	0.2058	1.3499	0.2280	3.3201	0.1623
1.1	3.0042	0.3393	1.7333	0.2927	9.0250	0.4412
Sum	8.8538	1.0	5.9212	1.0	20.4555	1.0

Logit Selected = 3

Generated Image:

2	3	3	3
3	3	3	2

3	3	3	3
3	3	3	$?_{44}$

Pixel [4,4]

Image Patch: Right boundary padding zeros out sum

Convolution result: $(3\times1)+(3\times-3)+(0)+(3\times2) = 0$

h = ReLU = 0

Logits (z = w·h + b):

$z_0 = 0.2 \times 0 + 0.1 = 0.1$

$z_1 = 0.3 \times 0 - 0.2 = -0.2$

$z_2 = 0.1 \times 0 + 0.3 = 0.3$

$z_3 = 0.4 \times 0 - 0.1 = -0.1$

T / Logits	exp(Li/T) T=1	Prob T=1	exp(Li/T) T=2	Prob T=2	exp(Li/T) T=0.5	Prob T=0.5
0.1	1.1052	0.2645	1.0513	0.2516	1.2214	0.2923
-0.2	0.8187	0.1959	0.9048	0.2165	0.6703	0.1604
0.3	1.3499	0.3230	1.1618	0.2780	1.8221	0.4361
-0.1	0.9048	0.2165	0.9512	0.2276	0.8187	0.1959
Sum	4.1786	1.0	4.0692	1.0	4.5326	1.0

Logit Selected = 2

Generated Image:

2	3	3	3
3	3	3	2
3	3	3	3
3	3	3	2

Final Generated Image

After processing all 16 pixels using the PixelCNN-style autoregressive generation:

2	3	3	3
3	3	3	2

3	3	3	3
3	3	3	2